VT-AC COMMANDS

A.C. WRIGHT DESIGN
VERSION 2.0.0 · NATIVE PERSONALITY

CONTROL CODES · $00–$1F

HEX	DEC	DESCRIPTION	STD ASCII	# BYTES	NOTES
00	0	NULL	Yes	1	No operation
01	1	CURSOR HOME	Yes	1	Move cursor to 0,0
02	2	CURSOR CHARACTER		2	2nd byte is the cursor character, or $00 to turn off `DEFAULT OFF`
03	3	CURSOR MODE		1	Toggle solid / blinking `DEFAULT SOLID`
04	4	RESET		1	Text mode, clear screen, cursor home / off / solid, BG $00, FG $FF
05	5	BELL DURATION		2	2nd byte is duration in jiffies (1⁄60 s) `DEFAULT $3C`
06	6	BELL FREQUENCY		2	2nd byte is a note index — see Bell Frequencies `DEFAULT $3D`
07	7	BELL	Yes	1	Queue one bell at the current duration and frequency
08	8	BACKSPACE	Yes	1	Cursor left one column; stops at column 0
09	9	TAB	Yes	1	Advance to the next 4-column stop
0A	10	LINE FEED	Yes	1	Cursor down, same column; scrolls up at the last row
0B	11	SCREEN MODE		1	Toggle text / graphics `DEFAULT TEXT`
0C	12	CLEAR SCREEN	Yes	1	Fill the screen with the background color
0D	13	CARRIAGE RETURN	Yes	1	Cursor to column 0, same row
0E	14	SET COLUMN		2	2nd byte is the column, taken modulo the column count — 0–39 at 40, 0–79 at 80 `DEFAULT 0`
0F	15	SET ROW		2	2nd byte is the row, taken modulo the row count — 0–29 at 40, 0–59 at 80 `DEFAULT 0`
10	16	DELETE TO SOL		1	Clear from the start of the line through the cursor
11	17	DELETE TO EOL		1	Clear from the cursor to the end of the line
12	18	DELETE TO SOS		1	Clear from the start of the screen through the cursor
13	19	DELETE TO EOS		1	Clear from the cursor to the end of the screen
14	20	SCROLL LEFT		1	Vacated column filled with the background color
15	21	SCROLL RIGHT		1	Vacated column filled with the background color
16	22	SCROLL UP		1	Vacated row filled with the background color
17	23	SCROLL DOWN		1	Vacated row filled with the background color
18	24	FOREGROUND COLOR		2	2nd byte is RGB332 color data `DEFAULT $FF WHITE`
19	25	BACKGROUND COLOR		2	2nd byte is RGB332 color data `DEFAULT $00 BLACK`
1A	26	NEXT BYTE AS DATA		2	Lets $00–$1F and $7F be interpreted as data rather than command
1B	27	ESC	Yes	2	Introduces a two-byte extension — see ESC Extensions
1C	28	CURSOR LEFT		1	Stops at column 0
1D	29	CURSOR RIGHT		1	Stops at the last column
1E	30	CURSOR UP		1	Stops at row 0
1F	31	CURSOR DOWN		1	Stops at the last row

CHARACTER DATA · $20–$FF

HEX	DEC	DESCRIPTION	STD ASCII	# BYTES	NOTES
20–7E	32–126	ASCII CHARACTERS	Yes	1	See ASCII chart
7F	127	DELETE	Yes	1	Clear the cell at the cursor
80–FF	128–255	EXTENDED CHARACTERS		1	CP437 upper range — see character set

ESC EXTENSIONS · $1B + 1 BYTE

v1.3.0 reserved `$1B` for future escape codes and treated it as a no-op. It is now the introducer for two-byte extensions. An unrecognised second byte is processed as an ordinary byte, so a stream that carried a stray ESC behaves exactly as it did under v1 — the deviation is the five sequences below and nothing else.

SEQUENCE	DESCRIPTION	# BYTES	NOTES
1B 01	**40-COLUMN MODE**	2	320 × 240 pixels, 40 × 30 cells. Clears the screen and homes the cursor `DEFAULT`
1B 02	**80-COLUMN MODE**	2	640 × 480 pixels, 80 × 60 cells. Clears the screen and homes the cursor
1B 03	**VT-100 PERSONALITY**	2	Hand the stream to the ANSI / VT-100 parser
1B 04	**QUERY**	2	Reply on the serial link with personality, columns and rows
1B 1B	**LITERAL ESC**	2	Emit `$1B` as data
1B *	**NO-OP ESC**	2	Any other second byte runs as if the ESC were not there — v1's behavior

The query reply is five bytes — `1B 04`, then personality (`00` native, `01` VT-100), then the column and row counts as literal numbers: `1B 04 00 28 1E` for 40 × 30, `1B 04 00 50 3C` for 80 × 60.

SCREEN GEOMETRY

MODE	PIXELS	CELLS	CELL
40-COLUMN	320 × 240	40 × 30	8 × 8
80-COLUMN	640 × 480	80 × 60	8 × 8

An exact 2× of the 40-column grid in both axes — same 8 × 8 font, square pixels, 4:3 aspect, and 8 rows per cell in graphics mode.

PERSONALITIES

MODE	PROTOCOL	ENTER
NATIVE	This card `DEFAULT`	ESC[?7000h
VT-100	ANSI / VT-100	1B 03

Native is byte-for-byte v1.3.0 plus the ESC extensions above. VT-100 mode is documented separately — see the README.

BELL FREQUENCIES · 2ND BYTE OF $06

Note index `$01`–`$54`, C1 through B7. Any value outside that range silences the bell, as does a duration of `$00`. Default is `$3D` — C6, 1046.50 Hz.

OCT	C	C♯	D	D♯	E	F	F♯	G	G♯	A	A♯	B
1	01	02	03	04	05	06	07	08	09	0A	0B	0C
2	0D	0E	0F	10	11	12	13	14	15	16	17	18
3	19	1A	1B	1C	1D	1E	1F	20	21	22	23	24
4	25	26	27	28	29	2A	2B	2C	2D	**2E**	2F	30
5	31	32	33	34	35	36	37	38	39	3A	3B	3C
6	**3D**	3E	3F	40	41	42	43	44	45	46	47	48
7	49	4A	4B	4C	4D	4E	4F	50	51	52	53	54

Highlighted: `$2E` A4 = 440.00 Hz · `$3D` C6 = 1046.50 Hz (default). Duration is in jiffies — `$3C` = 60 = one second.